

Yash Agrawal Full Stack Developer

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Summary

Full Stack Developer with 2+ years of expertise in designing and developing scalable web applications using modern technologies like React.js, Next.js, Node.js, and microservices architecture. Skilled in database optimization (PostgreSQL, MongoDB), cloud computing (AWS, Azure), and implementing CI/CD pipelines for seamless deployments. Strong focus on performance tuning, security best practices (OAuth 2.0, JWT), and real-time data processing. Experienced in Agile methodologies, A/B testing, and automation, ensuring efficient development workflows and high-quality software solutions that enhance user experience and business operations.

Technical Skills

- **Frontend:** React.js, Next.js, Redux, React Router, Material-UI, Tailwind CSS, D3.js, WebSockets
- **Backend:** Node.js, NestJS, Express.js, GraphQL, Django, Python, RESTful APIs, Kafka, Celery
- **Databases:** PostgreSQL, MongoDB, Redis, SQL, Query Optimization, Sharding, Indexing
- **Security:** OAuth 2.0, JWT Authentication, AES Encryption, Role-Based Access Control (RBAC), HTTPS
- **Cloud & DevOps:** AWS (ECS, Fargate, Lambda, API Gateway, CloudWatch), Microsoft Azure (AKS, Load Balancer, Monitor), Docker, Kubernetes
- **Testing & CI/CD:** Jest, Cypress, Postman, Selenium, PyTest, GitHub Actions, Jenkins
- **Agile & Tools:** Agile, Scrum, A/B Testing, Jira, Git, CI/CD Pipelines

Professional Experience

- Full Stack Developer, Mercury** 08/2024 – Present | Remote, USA
- Created and developed the Business Expense Management Platform, collaborating with stakeholders to gather requirements and implement Agile methodologies. Led sprint planning and A/B testing, optimizing workflows for startups and improving financial tracking by 45%.
 - Engineered a high-performance React.js and Next.js frontend, integrating AI-driven UI personalization, interactive D3.js data visualizations, and WebSocket-based real-time expense updates, improving user visit and reducing load times by 50%.
 - Built a scalable Node.js backend with NestJS and microservices architecture, leveraging GraphQL, Kafka event streaming, and Redis caching, enabling seamless transaction processing for 1M+ concurrent users with 99.98% uptime.
 - Optimized PostgreSQL and MongoDB databases, using superior indexing, sharding, and query parallelization, easing query latency by 70%. Used Redis caching for financial reports, enhancing dashboard responsiveness for real-time analytics.
 - Ensured enterprise-grade security with OAuth 2.0, JWT authentication, and AWS Cognito. Developed CI/CD pipelines with GitHub Actions, Jest, Cypress, and Postman, reducing deployment failures by 75% and increasing test coverage.
 - Deployed on AWS (ECS, Fargate, API Gateway, and Lambda) with predictive auto-scaling, fault-tolerant design, and zero-downtime deployments. Integrated CloudWatch for real-time monitoring, lowering downtime by 40% during peak hours.

- Associate Full Stack Developer, Wipro** 01/2021 – 07/2022 | Karnataka, India
- Architected a scalable, multi-tenant e-commerce platform with Agile methodology, collaborating across teams, implementing A/B testing for feature validation, and ensuring smooth deployment and scalability.
 - Developed advanced frontend UI/UX using React, Redux, React Router for dynamic navigation, and Material-UI for responsive designs, focusing on user-centric interactions and seamless cross-device compatibility.
 - Optimized PostgreSQL database performance by creating composite indexes, fine-tuning SQL queries, leveraging table partitioning, and using connection pooling for faster data retrieval and reduced latency.
 - Built robust backend features using Python and Django, implementing secure JWT authentication, optimizing RESTful APIs, managing asynchronous tasks with Celery, and ensuring high-concurrency handling.
 - Implemented strong security measures including AES encryption for sensitive data, OAuth 2.0 for secure user authentication, HTTPS for data transmission, and RBAC using Django's built-in authorization framework.
 - Automated testing with PyTest, Selenium WebDriver, and Django TestCase for unit, integration, and end-to-end testing, ensuring code quality, bug-free deployments, and seamless CI/CD pipelines.
 - Deployed the platform on Microsoft Azure, leveraging Azure Kubernetes Service for container orchestration, Azure App Services for scalable deployments, Azure Load Balancer for traffic distribution, and Azure Monitor for real-time diagnostics.

Education

- Master of Science in Computer Science** 08/2022 – 05/2024 | North Carolina, USA
University of North Carolina at Charlotte
- Bachelor of Engineering in Information Science** 08/2018 – 07/2022 | Karnataka, India
Visvesvaraya Technological University